

THIS IS A SOURCES SOUGHT NOTICE ONLY FOR MARKET RESEARCH PURPOSES.

THIS IS NOT A REQUEST FOR PROPOSAL (RFP), REQUEST FOR QUOTATIONS (RFQ) OR INVITATION FOR BID (IFB).

NO FORMAL SOLICITATION IS BEING REQUESTED UNDER THIS NOTICE.

This Sources Sought Notice is for market research purposes to identify parties having an interest in and resources to support this requirement for an Energy Dispersive Detector upgrade to Oxford Instruments Ultim Infinity 170 for the United States Geological Survey (USGS), Geology, Geophysics, and Geochemistry Science Center (GGGSC) Denver, CO supporting the Mineral Resource Mission Area.

The result of this market research will contribute to determining the method of procurement. The Government will review the information requested in this announcement based on the responses to sources sought notice/market research, this requirement may be set-aside for small businesses or procured through full and open competition.

The North American Industry Classification System (NAICS) code is 334516 (Analytical Laboratory Instrument Manufacturing) and the size standard is 1,000 employees.

NOTE 1: The Buy American Act (41 U.S.C. 8301-05) is the primary federal procurement law providing a preference for domestic goods and manufactured products. As such, the USGS is seeking information on sources that can supply the described instrument either mined, produced or manufactured in the U.S. in sufficient and reasonably available commercial quantities of satisfactory quality.

A. Introduction:

The United States Geological Survey (USGS), Geology, Geophysics, and Geochemistry Science Center in Denver, CO, supporting the Energy and Minerals Mission Area has a requirement to purchase a new Energy Dispersive Detector brand name or equal to Oxford Instruments Ultim Infinity 170.

B. Background:

The Denver Microbeam Laboratory in the Geology, Geophysics, and Geochemistry Science Center houses several microanalytical pieces of instrumentation which are used to study critical minerals and energy resources and contribute to geologic mapping. The energy dispersive spectrometer (EDS) allows for in situ collection of chemical data at the micrometer scale. This imaging and chemical resolution provided by the EDS allows us to determine the deportment of critical minerals in potential new resources, study the changing fluid chemistries from which ore deposits form, and determine the paragenesis and relationships of accessory minerals used in geochronology studies. The work in the lab is the critical first step many scientists rely on to guide further downstream analyses.

C. Minimum Salient Characteristics for an Energy Dispersive Detector:

1. Physical Characteristics:

- a. Sensor size $\geq 150 \text{ mm}^2$
- b. Attachment to the existing FEI 450 Quanta FESEM rear EDS port through vendor supplied flange
- c. Motorized detector digitally controlled through software to fully insert and retract the detector
- d. Detector must have a thin window to allow for Be detection.
- e. Detector must have a built in Peltier cooling system

2. Functional Characteristics:

- a. Detector must be able to detect the elements Be through U.
- b. Detector must work with the Oxford E1 and X4 control boxes and seamlessly integrate with existing Oxford Unity and C Nano detector hardware.
- c. Detector and its operational software must be able to complete automated workflows to collect large areas and incorporate data collected simultaneously from existing Oxford Unity and C Nano detectors.
- d. Software must store the full EDS data cube of individual fields and stitched fields of large areas and allow for post extraction of user defined areas.
- e. Detector software must provide a rigorous quantification routine that allows user standard inputs and factory standard sets.
- f. Detector software must provide phase analysis.
- g. Detector software and hardware must have pulse pile up suppression and correction.

3. Performance Characteristics:

- a. The detector that will be installed has been individually characterized by the factory such that the displayed output matches the calculated spectrum with no peak shifts or gaps in the data.
- b. Detector resolution at C Ka $\leq 50 \text{ eV}$ at 50,000 cps throughput.
- c. Detector resolution at Mn Ka $\leq 130 \text{ eV}$ at 200,000 cps throughput.
- d. Detector resolution must not drift more than 2 eV from 1000 to 100,000 cps throughput.
- e. Detector can achieve 1,000,000 cps in the total spectrum with no observable detector artifacts including pulse pile up peaks.

(End of Minimum Salient Characteristics)

Businesses with the capability to meet these requirements must submit the following information:

1. Brand Name or Equal Product Information:

Identify the proposed product/manufacture meeting the salient characteristics above. Include specification sheets and brochures.

2. Place of Manufacturing / Country of Origin:

Specify whether the product is manufactured within the United States in accordance with **NOTE 1**.

3. Business Size and Socioeconomic Status applicable to the NAICS code listed above:

Must indicate whether your business is:

- Indian Small Business Economic Enterprise (ISBEE)
- Small Business
- 8(a)
- HUBZone
- Service-Disabled Veteran-Owned Small Business (SDVOSB)
- Other than Small Business

4. Company Information:

Provide a point of contact (name and email), System for Award (SAM) Unique Entity ID (UEI), and confirmation of active SAM.gov registration.

All information submitted in response to this announcement is voluntary – the Government will not pay for information requested nor will it compensate any respondent for any cost incurred in developing information provided to the Government. Materials submitted to the Government for review will not be returned. Respondents will not be notified of the results of the evaluation.

This is NOT a Request for Proposal (RFP), Request for Quotations (RFQ) or Invitation for Bid (IFB). The Government does not intend to award a contract based on responses received under this announcement.

The government will evaluate market information to ascertain potential market capacity to provide services consistent in scope and scale with those described in this notice and otherwise anticipated.

To be considered responsive to this Notice, businesses must submit all required documentation by May 11, 2026 , at 2:30 p.m. Mountain Daylight Time (MDT). Any response that does not follow these instructions will not be considered.

Telephone inquiries will not be accepted or acknowledged, and no feedback or evaluations will be provided to companies regarding their submissions.

Inquiries will only be accepted by e-mail to Lisa Williams at lisa_williams@ios.doi.gov.